Module 2
System Installation and System Software

System Installation

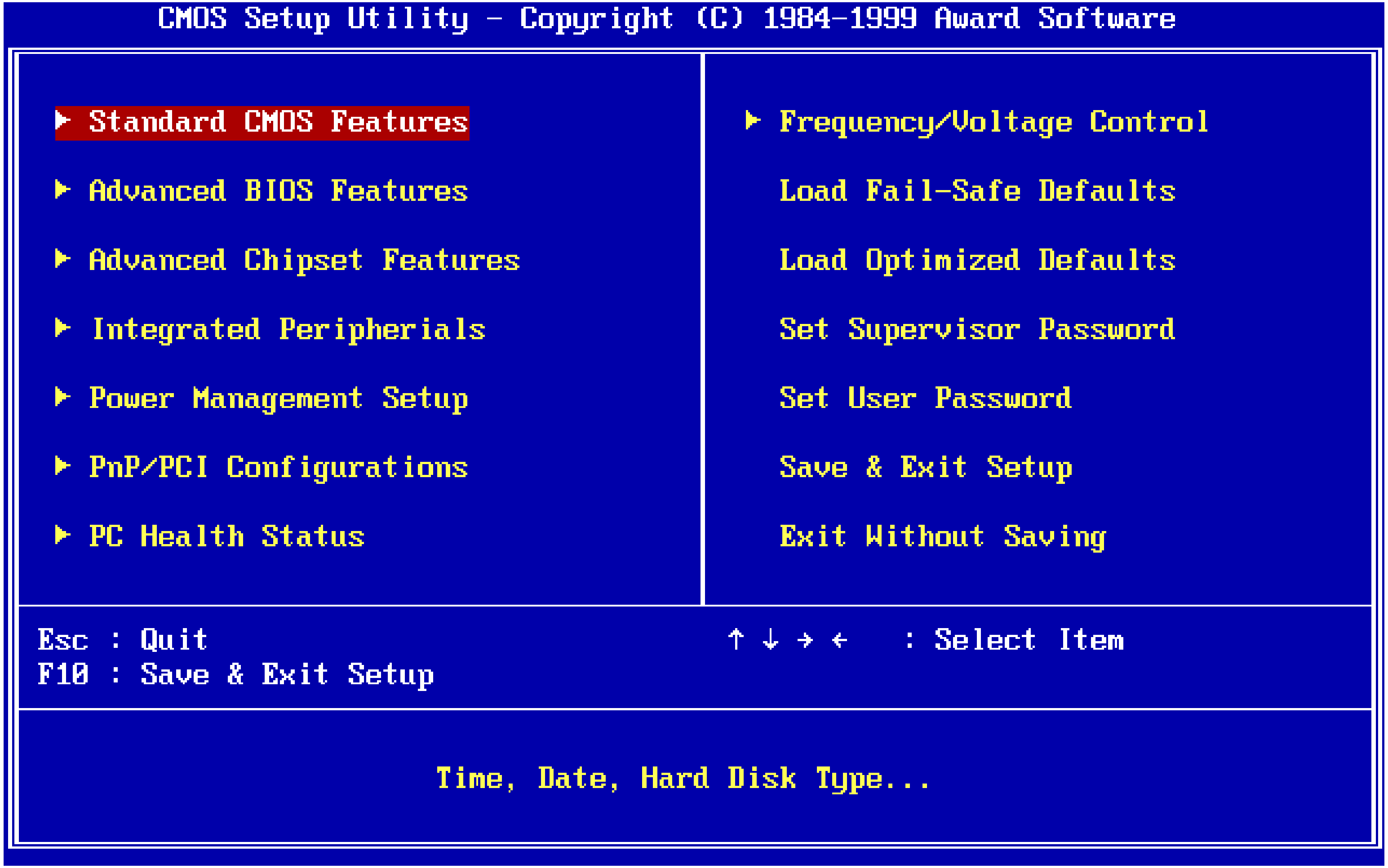
- The process of setting up, configuring, and preparing hardware or software to operate within a specific environment.
  - copying necessary files
  - establishing system connections
  - adjusting settings

System Software

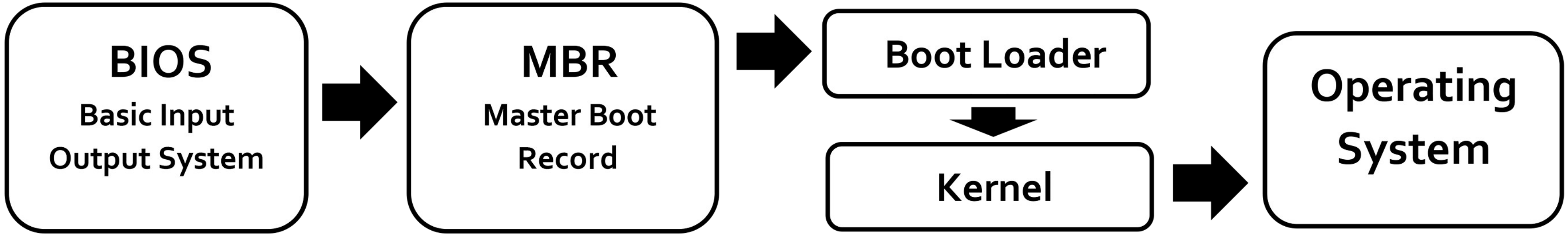
- Software that provides a platform for other software. Examples include:
  - Firmware
  - Utility programs
  - Operating systems
  - Device drivers
  - Hypervisors

Basic Input/Output System (BIOS)

- A type of firmware used to provide runtime services for OSes and programs.
- BIOS comes pre-installed on the computer's motherboard.
- BIOS performs hardware initialization during the booting process (power-on startup).


CMOS Setup Utility - Copyright (C) 1984-1999 Award Software
Standard CMOS Features
Advanced BIOS Features
Advanced Chipset Features
Integrated Peripherials
Power Management Setup
PnP/PCI Configurations
PC Health Status
Frequency/Voltage Control
Load Fail-Safe Defaults
Load Optimized Defaults
Set Supervisor Password
Set User Password
Save & Exit Setup
Exit Without Saving
Esc : Quit
F10 : Save & Exit Setup
↑ ↓ → ← : Select Item
Time, Date, Hard Disk Type...

Legacy Boot


BIOS
Basic Input
Output System
MBR
Master Boot
Record
Boot Loader
Kernel
Operating
System

For a picture to illustrate booting sequence:
https://en.wikipedia.org/wiki/Booting

Master Boot Record

- The MBR holds the information on how the disc's sectors are divided into partitions, each partition notionally containing a file system.
- The MBR also contains executable code to function as a loader for the installed OS.
- This MBR code is usually referred to as a boot loader, such as GNU GRUB, BOOTMGR

BIOS limitations (1)

- BIOS relies on the MBR partitioning scheme.
- MBR uses 32-bit values to store sector counts, which caps its maximum addressable storage space at 2.2 Terabytes (TB).
- MBR only supports a maximum of 4 primary partitions.

BIOS limitations (2)

- BIOS runs in 16-bit real mode. It can only address 1 Megabyte (MB) of system memory.
- Because of this, it cannot easily initialize multiple hardware peripherals simultaneously, leading to slower boot times.

BIOS limitations (3)

- BIOS executes any code found in the boot sector of the drive without verification.
- This made it highly vulnerable to rootkits and bootkits—malware that loads before the operating system and antivirus software even start, making them nearly impossible to detect.

Unified Extensible Firmware Interface (UEFI)

- BIOS limitations had become too restrictive for the larger server platforms.
- UEFI is a specification for the firmware architecture to overcome BIOS limitations.
- UEFI works with the GUID Partition Table (GPT) partitioning scheme, which is free from many of the limitations of MBR.
- UEFI supports 32-bit or higher processor architectures.
- Only processors with a little-endian mode are supported.

UEFI Secure Boot

- A feature in UEFI firmware that verifies the digital signature of the bootloader against a trusted list of keys embedded in the firmware.
- If the code has been tampered with or is from an unverified source, Secure Boot stops the process.
- UEFI is supported by Microsoft Windows and most modern Linux distros.

Power-On Self-Test (POST)

- Both BIOS and UEFI perform power-on self-test before a boot loader load an OS.
- BIOSes commonly use a sequence of beeps from the motherboard-attached PC speaker (if present and working) to signal error codes.
- POST beep codes vary from manufacturer to manufacturer.

Good-to-know Utility Programs

- GParted – Disk partition management
- Clonezilla – Disk cloning and imaging
- Memtest86+ – Memory test
- TestDisk – Disk test

How to use these tools is out of scope...

10 minutes
Discussion

What are common hardware problems
that you have encountered in the past?

For example: disk failure,
corrupted boot partition, etc

Could you identify the issues by yourself?
How did you confirm your hypotheses?

Operating Systems

- Personal computers (Desktop & Laptop)
- Mobile phones & Tablets
- Servers

Well-known Operating Systems

- Personal Computer
  - Microsoft Windows
  - macOS
  - Chrome OS
  - Linux
- Mobile devices
  - Android
  - Apple iOS

Server Operating Systems

- Unix: IBM AIX, HP-UX, Oracle Solaris
- BSD: FreeBSD, NetBSD, OpenBSD, etc.
- Linux: Red Hat Enterprise Linux, Fedora, Debian, Ubuntu
- Microsoft Windows Server

Unix

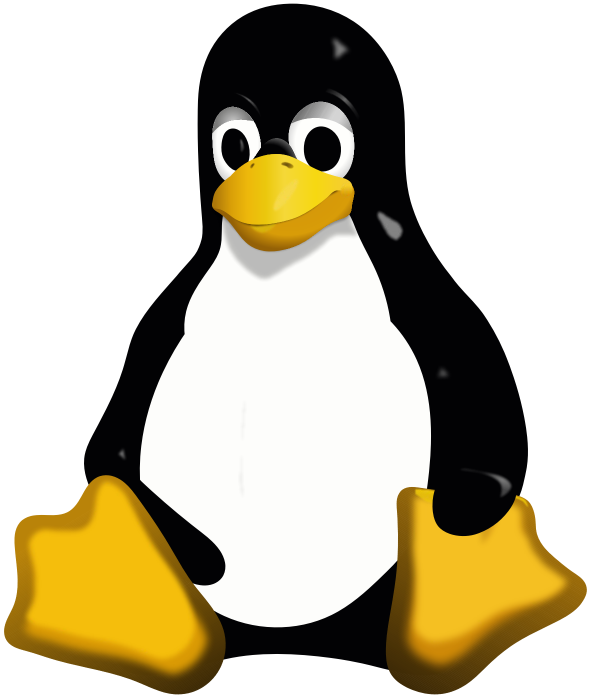
- Family of multitasking, multi-user computer operating systems
- Derived from the original AT&T Unix
- Original Unix was developed in 1969 at the Bell Labs research center.

Berkeley Software Distribution (BSD)

- First released in 1978, it began as an improved derivative of AT&T's original Unix developed at Bell Labs.
- Berkeley ended its Unix research in 1992.
- The term "BSD" came to refer primarily to its open-source descendants, including FreeBSD, OpenBSD, NetBSD, etc.

Linux

- Family of open-source software Unix-like OSes based on the Linux kernel.
- The Linux kernel was created by Linus Torvalds.



Many thousands of Linux distributions (distros)

Microsoft Windows Servers

- Microsoft Windows but for servers.
- Great for managing multiple Windows desktops & laptops for enterprises

- Main releases:
  - Windows Server 2025
  - Windows Server 2022
  - Windows Server 2016
  - Windows Server 2012
  - Windows Server 2008
  - …

Hypervisors

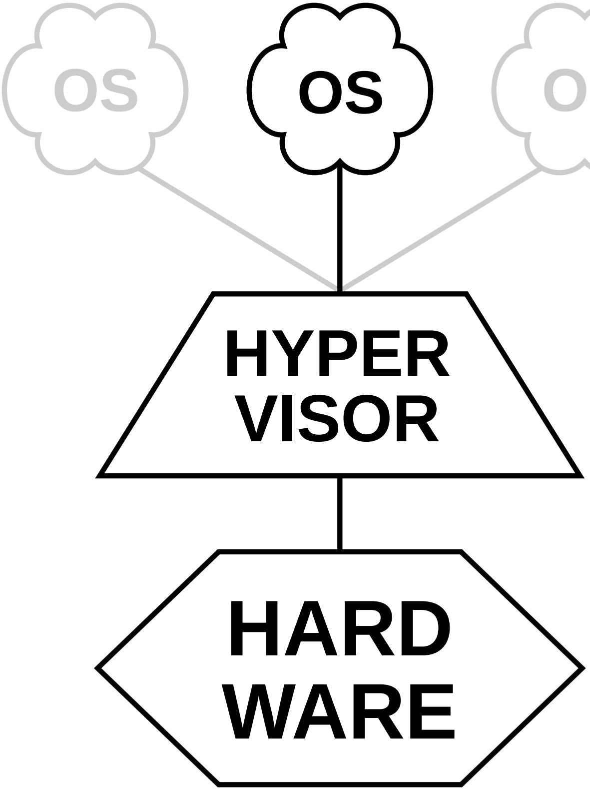
- A specialized software layer that enables multiple isolated virtual machines (VMs) to share a single physical host computer.
- Also known as virtual machine monitor.
- Ensures VMs cannot interfere with one another.

Type 1 Hypervisor (Bare-Metal)

- These hypervisors run directly on the host's hardware to control the hardware and to manage guest operating systems.
- They are sometimes called bare-metal hypervisors.
- Examples include VMware ESXi, Xen

Type 2 Hypervisor (Hosted)

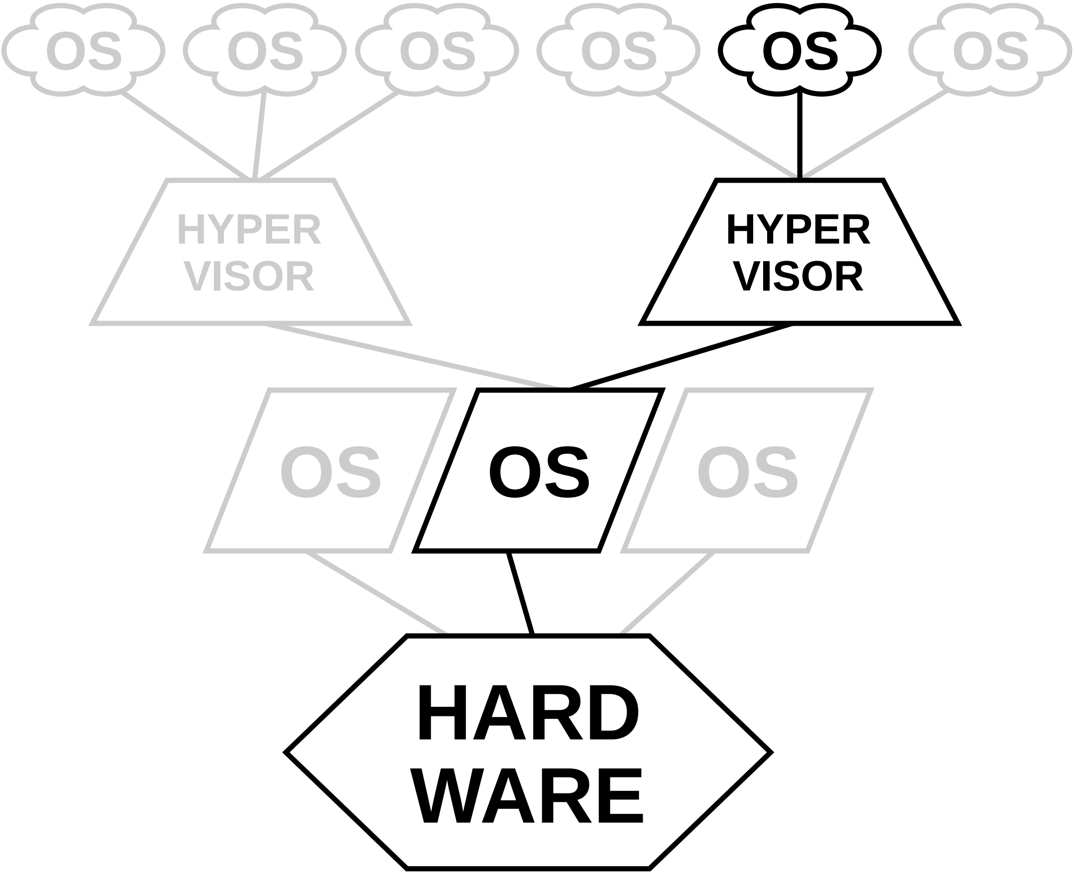
- These hypervisors run on a conventional operating system (OS) just as other computer programs do.
- Type-2 hypervisors abstract guest operating systems from the host operating system, effectively creating an isolated system that can be interacted with by the host.
- Examples include VirtualBox and VMware Workstation.



TYPE 1

native

(bare metal)



TYPE 2

hosted

It’s time to get your hands dirty!

1) Install on your machine a type-2 hypervisor, either VirtualBox or Vmware Workstation!
2) Create a virtual machine with 1 vCPU, 1.5GB of memory, and 6GB of disk.
3) Install Ubuntu Server 26.04 LTS on your VM!

- You may try Alpine Linux if your machine doesn’t have enough resources. 512MB of memory is OK!